Cambridge
O Level

**Cambridge International Examinations**
Cambridge Ordinary Level

**MATHEMATICS (SYLLABUS D)** **4024/21**

Paper 2 **May/June 2016**

MARK SCHEME

Maximum Mark: 100

**Published**

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| Question | Answers | Mark | Part Marks |
|---|---|---|---|
| **1 (a)** | 7.5(0) | **2** | M1 for $x + \frac{60x}{100} = 12$ soi or<br>B1 for ÷ by 160 |
| **(b)** | 45 | **2** | M1 for $\frac{17.40-12}{12}\times 100$ |
| **(c)** | 35 | **2** | M1 for $\frac{17.4-11.31}{17.4}\times 100$ |
| **(d)** | 25 | **3** | M1 for $60 \times 17.4 + x \times 11.31 (\geqslant 1320)$ or<br>B1 276<br>A1 for 24.4(03...) |
| **2 (a)** | 6 | **2** | M1 for $p - 1 = 5(7 - p)$ soi |
| **(b)** | $\frac{3b^2}{a}$ | **2** | M1 for $\frac{9b^4}{a^2}$ oe<br>$\frac{3a^{\frac{1}{2}}b^3}{a^{\frac{3}{2}}b}$ oe<br>or B1 for $3b^2$ as numerator or $\frac{k}{a}$ |
| **(c)** | $\frac{q^2}{3}$ | **2** | B1 for $q^2(1-q)$ or $3(1-q)$ |
| **(d) (i)** | $(4t-1)(t+9)$ | **2** | B1 for $(at+c)(bt+d)$<br>with $ab = 4$ or $cd = -9$ |
| **(ii)** | $\frac{1}{4} - 9$ or ft | **1ft** | |
| **3 (a)** | Correct graph | **2** | B1 for correct scales and 4 points or<br>wrong scales and all points. |
| **(b) (i)** | $-2.3 \pm 0.5$ $1.3 \pm 0.5$ | **1** | |
| **(ii)** | $-2.8 \pm 0.5$ $1.8 \pm 0.5$ | **2** | M1 for $x^2 + x - 3 = 2$ soi |
| **(c)** | 2.4 to 3.6 | **2** | M1 for tangent at $x = 1$ |
| **(d) (i)** | $y = 2x - 2$ | **2** | B1 for $2x$ or $-2$ |
| **(ii)** | $-0.6$ 1.6 | **2** | Dependent on line drawn<br>B1 for their line having FT gradient or FT intercept |

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|---|---|---|---|
| **4 (a)** | Complete proof | **3** | B2 for 2 pairs of equal angles<br>1 pair with reason.<br><br>B1 for 1 pair of equal angles. |
| **(b) (i)** | 2 : 5 | **2** | B1 for $NM : BL = 2 : 3$ oe or $NM = LC$ |
| **(ii)** | 4 : 9 | **1** | |
| **(iii)** | 1 : 3 | **2** | B1 for such as<br>$\frac{\Delta ANM}{\Delta ABC} = \frac{4}{25}$ or<br>$\frac{\Delta NBL}{\Delta ABC} = \frac{9}{25}$ |
| **5 (a)** | 15.1 or 15.08(….. | **3** | M1 for $\tan\theta = \frac{31}{115}$ or $\tan\theta = \frac{115}{31}$<br><br>A1 for $\theta = 15.1$ or $\theta = 74.9$ |
| **(b) (i)** | 18.8 or 18.77…… | **2** | M1 for $\sin\theta = \frac{354}{1100}$ |
| **(ii)** | 251 or 251.2(……. | **1ft** | $270 -$ their $L\hat{J}K$ final ans. |
| **6 (a)** | $\begin{pmatrix} 6 & -2 \\ -5 & 11 \end{pmatrix}$ | **2** | B1 for at least 2 elements correct in a 2 x 2 matrix |
| **(b)** | $\begin{pmatrix} 15 & -7 \\ 7 & 8 \end{pmatrix}$ | **2** | B1 for at least 2 elements correct or<br><br>M1 for $\begin{pmatrix} 4 & -1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 4 & -1 \\ 1 & 3 \end{pmatrix}$ soi |
| **(c)** | $-\frac{1}{10}\begin{pmatrix} -5 & 0 \\ -7 & 2 \end{pmatrix}$ oe<br><br>isw | **2** | B1 for det B = $-10$ soi or<br>$\begin{pmatrix} -5 & 0 \\ -7 & 2 \end{pmatrix}$ |
| **(d)** | $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ | **1** | |
| **(e)** | $\begin{pmatrix} 0 & 0 \\ 7 & -7 \end{pmatrix}$ | **2** | B1 for $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ soi |

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|---|---|---|---|
| 7 (a) | 4.53 to 4.54 | 4 | B2 for BOC = 52 or after B0<br><br>B1 for $A\hat{B}C = 90$ or triangle $OBC$ isosceles or $B\hat{A}C = 26$<br><br>M1 for $\frac{52}{360} \times 2\pi 5$ ft |
| (b) (i) | 101 or 32π or 100 to 100.6 | 2 | M1 for $\pi(16.52\ )$ or $15.5^2$ |
| (ii) | 0.87 to 0.871 | 3 | B1 for $\pi 15.5^2$ or $44\pi r^2$ and<br><br>M1 for $r^2 = \frac{\pi 15.5^2 - 650}{44\pi}$ |
| (iii) | 7 | 3 | M1 for $\pi 15.5^2 d = 500$<br><br>A1 for 0.66 to 0.663 |
| 8 (a) (i) | $-1.92$ **(3……** | 1 | |
| (ii) | $\frac{8}{p+5}$ | 2 | M1 for $\frac{8}{q} = p + 5$ or $pq = 8 - 5q$ or $p = \frac{8}{q} - 5$ |
| (b) (i) | $H$ and $h$ correctly derived | 2 | M1 for correct substitution in the formula for the area of a trapezium. |
| (ii) | $\frac{75}{(x-1)(2x+3)}$ correctly derived | 3 | M1 for $\frac{15(2x+3) - 30(x-1)}{(x-1)(2x+3)}$ soi<br><br>B1 for $30x + 45 - 30x + 30$ soi |
| (iii) (a) | Equation correctly derived. | 2 | B1 for $\frac{75}{(x-1)(2x+3)} = 1.5$ |
| (b) | 4.90 | 2 | B1 for $\sqrt{1^2 - 4 \times 2 \times (-53)}$ soi or<br><br>B1 for $\frac{-1 \pm \sqrt{their\,425}}{2 \times 2}$ soi |

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|---|---|---|---|
| 9 (a) (i) | 5.38 to 5.39 or √29 | 2 | M1 for $(AC^2) = 2^2 + 5^2$ |
| (ii) | 0.517 to 0.518 | 2 | M1 for $\frac{CE}{2} = \sin 15$ oe |
| (iii) | 68.8 to 68.9 | 4 | M1 for $\frac{AF}{2} = \cos 15$ oe or $BC^2=BE^2+$ (their CE)$^2$ or any complete alternative method<br>A1 for 1.932 and<br>M1 for $\tan F\hat{A}E = \frac{5}{2\cos 15}$ oe or $\frac{5}{their(AF)}$ |
| (b) (i) | 80.9(4…. Or 81 | 3 | B1 for $10^2 = 6^2 + 9^2 - 2\times6\times9\times\cos\theta$<br>or<br>B2 for $\cos\theta = \frac{9^2 + 6^2 - 10^2}{2\times9\times6}$ |
| (ii) | > | 1 | |
| 10 (a) | (2) (4) 14 54 84 98 (100) | 1 | |
| (b) | Correct curve | 2 | P1 for at least 5 correct plots |
| (c) (i) | 195 ft 190 ⩽ and < 200 | 1 | |
| (ii) | 50 –75 | 2 | B1 for one quartile correct in ranges 225 to 235 or 160 to 175 |
| (d) | Correct curve | 4 | P3 for at least 4 correct plots or<br>B1 + B1 for any two correct points soi. |
| (e) | 92 ft | 1 | |
| (f) | B 15 ft A | 1ft | Their 90 – 75 |

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|---|---|---|---|
| **11 (a)** | $\begin{pmatrix}-6\\2\end{pmatrix}$ | **1** | |
| **(b) (i)** | $\begin{pmatrix}8\\4\end{pmatrix}$ | **2** | B1 for $\begin{pmatrix}8\\k\end{pmatrix}$ or $\begin{pmatrix}k\\4\end{pmatrix}$ |
| **(ii)** | $\begin{pmatrix}-8\\-4\end{pmatrix}$ ft | **1** | |
| **(iii)** | 8.94 or 8.94 to 8.95<br>or √80 oe | **2** | M1 for $\sqrt{(-8)^2+(-4)^2}$ oe ft |
| **(c) (i)** | Triangle vertices (5,4), (13,0), (9,8) | **2** | B1 for 2 correct |
| **(ii)** | Triangle *F* (5,4), (7,3), (6,5) | **1** | |
| **(iii)** | Rotation<br>180<br>Centre (5,4) | **3** | B2 for Rotation with either centre or angle.<br><br>B1 for Rotation. |

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